Panskura Banamali College (Autonomous)

B.Sc.(H) 6th Semester Examination -2022 (Online Mode)

Subject: BCA Paper: CC14T

Paper Name: Theory of Automata and Compiler Design

F.M.: 40 Time : 2 Hours

Answer any four questions: 4x10=40

1. Construct the DAG for following statement. a+b*c+d+b*c. 10

2. Explain the simple code generator and generate target code sequence for the following statement d:=(a-b)+(a-c)+(a-c) . 10

3. Consider the grammar
   S->AB|Abad
   A->d
   E ->b
   D->b| ε
   B->c
   Construct the predictive parsing table. Show that the given grammar is LL(1) or not. 10

4. a) Construct the recursive decent parser for the following grammar?
   E-> E+T/T
   T-> T*F/F
   F-> (E)/id
   b) Explain about Left factoring and Left Recursion with an examples? 5+5=10

5. Define augmented grammar? Construct the LR(0) items for the following Grammar?

S->L=R

S->R

L->*R

L->id

R->L 10

6. Construct CLR Parsing table for the given grammar
   S->CC
   C->aC/d 10

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